

competitive but even outpaces the RTX 4060 by 10 per cent in raster performance while also delivering a 24 per cent improvement over its predecessor, the ARC A750.

However, NVIDIA maintains its edge in ray tracing. With 3rd-generation Ray Tracing cores and DLSS 3.5 technology, the RTX 4060 offers smoother visuals and higher frame rates in ray-traced titles. Its Frame Generation feature improves performance further, making it a strong choice for gamers invested in visually rich environments.

AMD's RX 7600, meanwhile, sticks to its strengths in rasterisation. While its ray tracing capabilities lag behind Intel and NVIDIA, its 32 Compute Units and Infinity Cache allow it to deliver reliable performance at 1080p and light 1440p gaming. It's a pragmatic choice for buyers who prioritise efficiency over cutting-edge features. Now, we will test and verify all these claims all over again once the Intel card shows up. For now, it does look like a good choice for the everyday gamer.

AI – The buzzword that is a reality now

In an age where AI-driven enhancements are becoming a must-have, Intel's XMMAI engines bring competitive features to the table. From image upscaling to generative tasks like real-time filters and creative workflows, Intel is positioning the ARC B580 as a tool not just for gamers but also for creators.

NVIDIA, however, remains the undisputed leader in this space. Its 4th-generation Tensor Cores power DLSS 3.5, enabling features like Frame Generation and Ray Reconstruction. These technologies elevate gaming visuals and make the RTX 4060 an excellent choice for users who want the best of AI-driven enhancements. AMD's approach to AI is more modest. The RX 7600's AI capabilities, while present, do not match the refinement or scope of NVIDIA and Intel. This reflects AMD's broader strategy of focusing on core performance rather than chasing AI-driven innovations.



ARC B580	ARC B570	
20	18	X ^e Cores
5	5	Render Slices
20	18	Ray Tracing Units
160	144	XMMAI Engines
2670 MHz	2500 MHz	Graphics Clock
12 GB	10 GB	Memory
192 bit	160 bit	Memory Interface
456 GB/s	380 GB/s	Memory Bandwidth
233	203	Peak TFLOPs (int. series)
190 W	150 W	Total Board Power
1x 8-Pin		Power Connector
PCIe 4.0 x8		PCIe Configuration
AV1, HEVC, AVC, VP9, XAVC-H		HW Accelerated Media
3x DisplayPort 2.1 (up to UHBR 13.5)		Display Configuration
1x HDMI 2.1		

What about the electricity bills?

Efficiency is a critical factor for Indian buyers, who are often mindful of electricity costs and long-term ownership expenses. The ARC B580's 190W power draw, while manageable, is notably higher than the RTX 4060's highly efficient 115W consumption. NVIDIA's ability to deliver performance at lower power levels strengthens its appeal, especially for users prioritising energy efficiency.

AMD's RX 7600 strikes a balance between the two, with a typical board power of 165W. It offers better efficiency than Intel without sacrificing much performance.

I want 1440p!

The ARC B580 comes with 12 GB of GDDR6 memory with a bandwidth of 456 GB/s, giving it a distinct advantage in memory-intensive scenarios. This makes it particularly suitable for games at 1440p and beyond, where larger textures and higher graphical settings demand more memory. And they claimed that this is a 1440p card. So, it will be interesting to see how it fares in our tests. Moving on.

In contrast, the RTX 4060 and RX 7600 are limited to 8 GB of GDDR6 on a 128-bit bus. While NVIDIA compensates for this with its memory management through Ada Lovelace architecture, and AMD uses its Infinity Cache to boost effective bandwidth, the gap in raw memory capacity may become a

bottleneck for these GPUs in memory-hungry games.

In the driving seat

Software support and driver reliability are make-or-break factors in the Indian market. NVIDIA's GeForce Experience/App suite is the gold standard, offering features like Ansel, Highlights, and Reflex. These tools enhance gaming and productivity workflows, making the RTX 4060 a well-rounded choice. AMD's Adrenalin Edition software provides similar functionality, with features like Radeon Boost and Anti-Lag aimed at competitive gamers. However, its driver support, while much improved, still lags behind NVIDIA's seamless updates.

Intel's ARC Control software is relatively new and offers promising features like overclocking controls and performance monitoring. However, driver maturity remains a question mark. Intel will need to address this quickly to build confidence. They have come a long way from where they started, having released over 50 drivers since launch and expanding support to 2.5 times the games that they did have support for. If they continue on this path, they might add a lot of appeal to their new card. **d**

